

A probability primer

- 1- S. Ross, A First Course in Probability, Pearson.
- 2- D. P. Bertsekas, Introduction to Probability, Athena Scientific.
◆
- 3- A. Papoulis, Probability, Random Variables and Stochastic Processes, McGraw Hill.

Why Probability ?

Why Probability ?

Deep learning is nothing else than probability. There are two principles involved in it, one is the maximum likelihood and the other one is Bayes. It is all about maximizing a likelihood function in order to find the probability distribution and parameters that best explain the data we are working with.



<https://medium.com/geekculture/an-introduction-to-probabilistic-deep-learning-explained-in-simple-terms-7c9d7005237b>

The French mathematician Laplace declared that probability theory is “common sense reduced to calculation.” In fact, it is just that - a way of numerically encoding our state of knowledge about variables in the world. Often the variables we care about are some measured data, and we wish to make inferences from this data in the face of uncertainty

- Prediction (Weather, Finance, Success in anything etc.)
- Estimation (Noisy data, Kalman Filter, Noise cancellation etc.)
- Decision making (Medical diagnosis etc.)
- Communication (Channel and source allocation etc.)
- Information (Coding etc.)

Probability is the branch of [mathematics](#) concerning numerical descriptions of how likely an [event](#) is to occur, or how likely it is that a proposition is true. The probability of an event is a number between 0 and 1, where, roughly speaking, 0 indicates impossibility of the event and 1 indicates certainty

Consider an experiment that can produce a number of results. The collection of all possible results is called the [sample space](#) of the experiment, sometimes denoted as Ω . The [power set](#) of the sample space is formed by considering all different collections of possible results.

A probability is a [way of assigning](#) every event a value between zero and one,

<https://en.wikipedia.org/wiki/Probability>

When dealing with [random experiments](#) – i.e., [experiments](#) that are [random](#) and [well-defined](#) – in a purely theoretical setting (like tossing a coin), probabilities can be numerically described by the number of desired outcomes, divided by the total number of all outcomes. This is referred to as **theoretical probability** (in contrast to [empirical probability](#), dealing with probabilities in the context of real experiments)

<https://en.wikipedia.org/wiki/Probability>

We say that the probability of an event A occurring is

$$P(A) = \frac{\text{Number of elements in } A}{\text{Total number of elements in the sample space}}$$

Example

If a fair coin is tossed, it is clear from our definition of probability above that $P(\text{obtaining a head}) = \frac{1}{2}$.

Example

A card is selected at random from a pack of 52 cards. Let $A =$ ‘the card is a heart’ and $B =$ ‘the card is an ace’.

Find $P(A)$, $P(B)$.

Solution

$P(A) = \frac{13}{52}$ since there are 13 hearts in the pack. $P(B) = \frac{4}{52}$ since there are 4 aces in the pack.

Random Variable

- **Random variable** is a quantity resulting from an experiment that, by chance, can assume different values.
- Experiment: Tossing a coin three times
- Random variable: the number of heads.
- Experiment: Rolling a dice.
- Random variable: the number appears face up.

Discrete probability

Axioms

The probability, P , that a discrete valued variable, X , occupies a specific state, x , is a number between zero and one:

$$0 \leq P(X = x) \leq 1. \quad (1)$$

The sum of the probabilities of all outcomes equals one:

$$\sum_x P(X = x) = 1. \quad (2)$$

From here on out we shall use the shorthand $P(x)$ to stand for $P(X = x)$.

PROBABILITY DISTRIBUTION A listing of all the outcomes of an experiment and the probability associated with each outcome.

Discrete Random Variables - Examples

- The number of students in a class.
- The number of children in a family.
- The number of cars entering a carwash in an hour.

Continuous Random Variables - Examples

- The distance students travel to class.
- The time it takes an executive to drive to work.
- The length of an afternoon nap.
- The length of time of a particular phone call.

Uniform

The uniform distribution is the most trivial form of distribution, where all outcomes are equally likely:

$$P(x) = 1/n, \quad x = 1, \dots, n. \quad (3)$$

Bernoulli

A Bernoulli random variable is simply a binary variable (i.e., it occupies one of two states) with

$$P(1) = p \quad (4)$$

$$P(0) = 1 - p \quad (5)$$

Binomial

If we draw n independent samples from a Bernoulli variable, the total number of 1's that we get is a binomial random variable.

The probability of any particular outcome with i one's and $n-i$ zero's is $p^i (1-p)^{(n-i)}$.

Since this can happen $\binom{n}{i}$ different ways, the total probability of getting i one's is thus given by equation 6.

$$P(i) = \binom{n}{i} p^i (1-p)^{(n-i)} . \quad (6)$$

where p is as specified in (4,5). The notation $\binom{n}{i}$ means “ n choose i .” It tells us the number of ways we could get i one's out of n draws:

$$\binom{n}{i} = \frac{n!}{(n-i)!i!} . \quad (7)$$

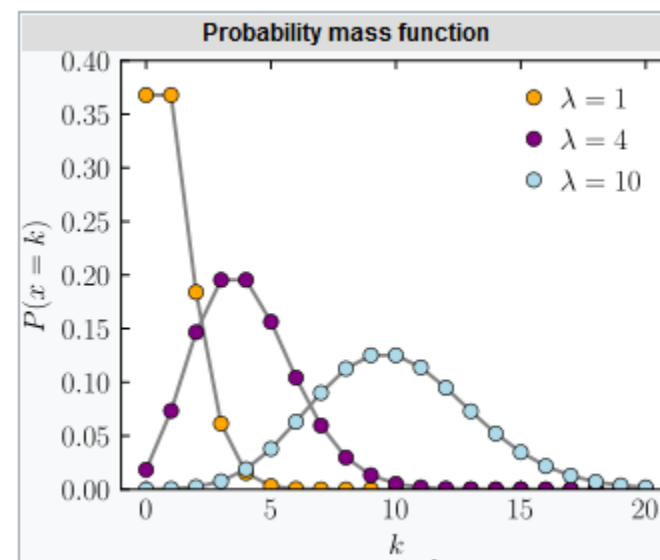
Poisson distribution

In probability theory and statistics, the **Poisson distribution** is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event.^[1] It is named after French mathematician Siméon Denis Poisson (/ˈpwɑːsɒn/; French pronunciation: [pwa.sɔ̃]). The Poisson distribution can also be used for the number of events in other specified interval types such as distance, area, or volume. It plays an important role for discrete-stable distributions.

For instance, a call center receives an average of 180 calls per hour, 24 hours a day. The calls are independent; receiving one does not change the probability of when the next one will arrive. The number of calls received during any minute has a Poisson probability distribution with mean 3. The most likely number of calls received are 2 and 3, but 1 and 4 are also likely. There is a small probability of it being as low as zero and a very small probability it could be 10 or even higher.

Another example is the number of decay events that occur from a radioactive source during a defined observation period.

$$P(i) = e^{-\lambda} \frac{\lambda^i}{i!}$$



https://en.wikipedia.org/wiki/Poisson_distribution

- : The number of misprints on a page of a book
- : The number of calls in a period of time
- : The number of decayed particles in a fixed period from radioactive material
- : The number of spikes discharged from a neuron in a period.

The Poisson distribution can be derived as a limiting case to the [binomial distribution](#) as the number of trials goes to infinity and the [expected](#) number of successes remains fixed — see [law of rare events](#) below. Therefore, it can be used as an approximation of the binomial distribution if n is sufficiently large and p is sufficiently small. The Poisson distribution is a good approximation of the binomial distribution if n is at least 20 and p is smaller than or equal to 0.05, and an excellent approximation if $n \geq 100$ and $np \leq 10$.^[30]

$$F_{\text{Binomial}}(k; n, p) \approx F_{\text{Poisson}}(k; \lambda = np)$$

https://en.wikipedia.org/wiki/Poisson_distribution

Joint distributions

The probability of two or more variables occupying a combined state, $X_1 = x_1$, $X_2 = x_2, \dots$ $X_n = x_n$, is denoted by $P(x_1, x_2, \dots, x_n)$ or $P(\mathbf{x})$. Such a distribution obeys the same axioms as above, $0 \leq P(\mathbf{x}) \leq 1$, and $\sum_{\mathbf{x}} P(\mathbf{x}) = 1$.

Conditional probability

We can express the interaction between variables using the conditional probability

$$P(x|y) = \frac{P(x, y)}{P(y)} \tag{9}$$

Definition 3.2

Consider two discrete random variables X and Y . We say that X and Y are independent if

$$P(X = x, Y = y) = P(X = x)P(Y = y), \quad \text{for all } x, y.$$

In general, if two random variables are independent, then you can write

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B), \quad \text{for all sets } A \text{ and } B.$$

$$P(x \mid y) = \frac{P(x,y)}{P(y)} = P(x)$$

Definition 3.3

Consider n discrete random variables $X_1, X_2, X_3, \dots, X_n$. We say that $X_1, X_2, X_3, \dots, X_n$ are independent if

$$\begin{aligned} &P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) \\ &= P(X_1 = x_1)P(X_2 = x_2) \dots P(X_n = x_n), \quad \text{for all } x_1, x_2, \dots, x_n. \end{aligned}$$

Suppose that X and Y have the following [joint probability mass function](#),^[6] in which the six central cells give the discrete joint probabilities $f(x, y)$ of the six hypothetical realizations $(x, y) \in S = \{(5, 8), (6, 8), (7, 8), (5, 9), (6, 9), (7, 9)\}$:

$f(x, y)$		x			$f_Y(y)$
		5	6	7	
y	8	0	0.4	0.1	0.5
	9	0.3	0	0.2	0.5
$f_X(x)$		0.3	0.4	0.3	1

$$f_Y(y) = \sum_x P(x, y)$$

$$f_X(x) = \sum_y P(x, y)$$

$$P(X = 5 | Y = 8) = \frac{P(X=5, y=8)}{P(Y=8)} = \frac{0}{0.5} = 0$$

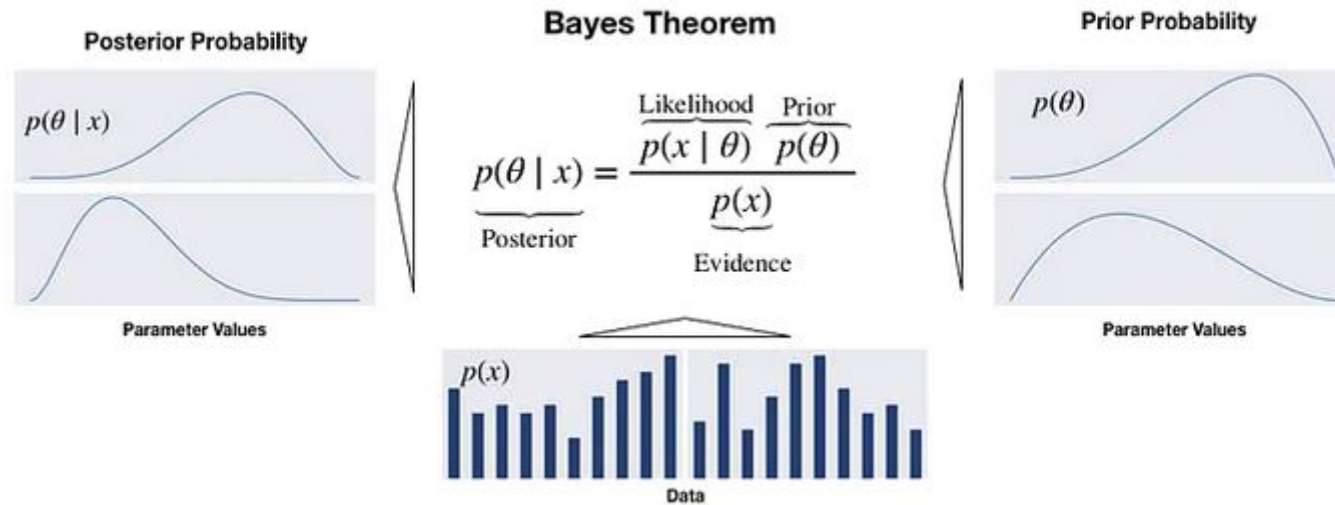
$$P(X = 6 | Y = 8) = \frac{P(X=6, y=8)}{P(Y=8)} = \frac{0.4}{0.5} = 0.8$$

$$P(X = 7 | Y = 8) = \frac{P(X=7, y=8)}{P(Y=8)} = \frac{0.1}{0.5} = 0.2$$

Bayes' theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where A and B are events and $P(B) \neq 0$.



<https://odsc.medium.com/how-bayesian-machine-learning-works-5fd1a746734>

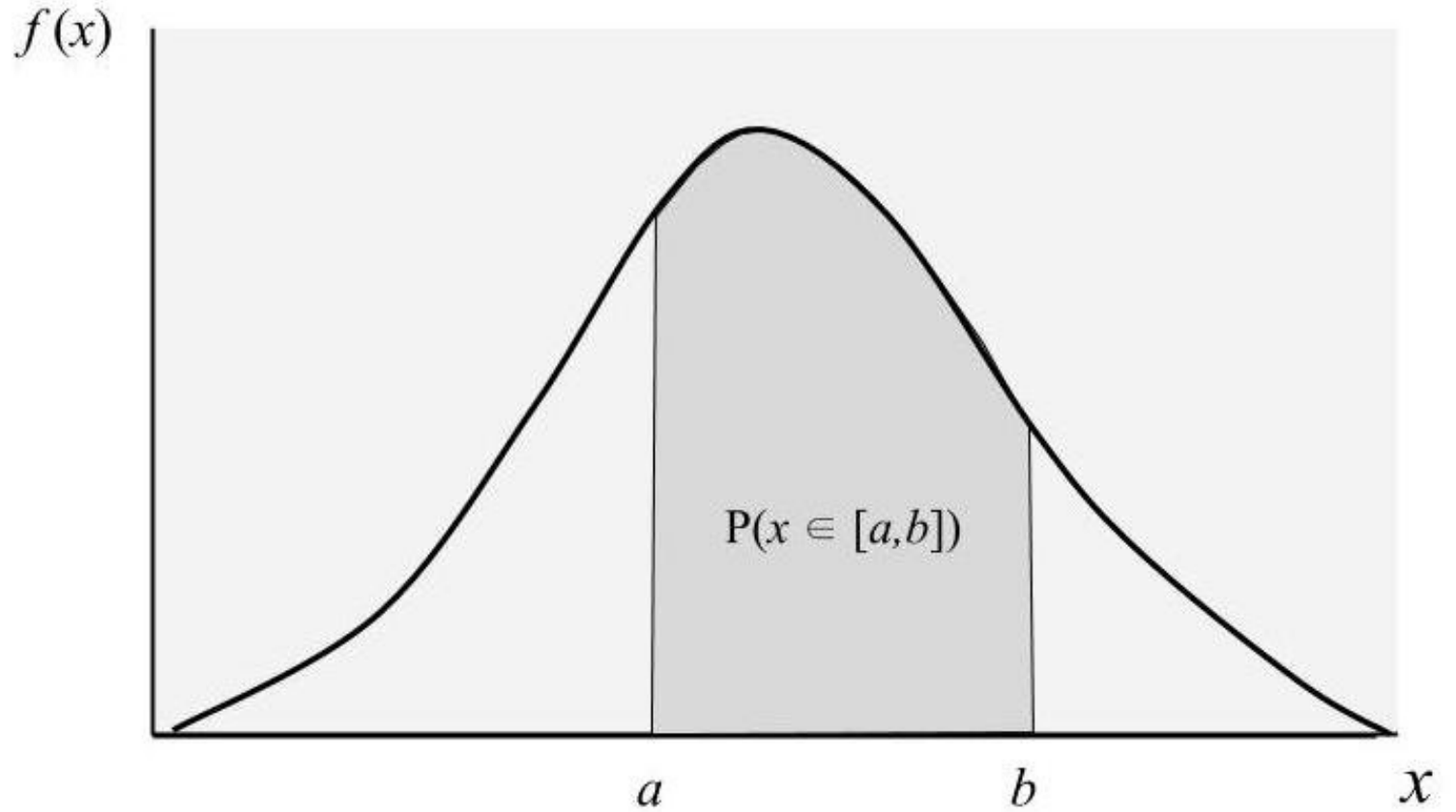
<https://machinelearningmastery.com/bayes-theorem-for-machine-learning/>

Continuous variables

Probability Density Function

$$p(x) \geq 0$$

$$\int_{-\infty}^{\infty} p(x) dx = 1$$



$$P[a \leq X \leq b] = P(x \in [a, b]) = \int_a^b f_X(x) dx$$

Cumulative distribution function

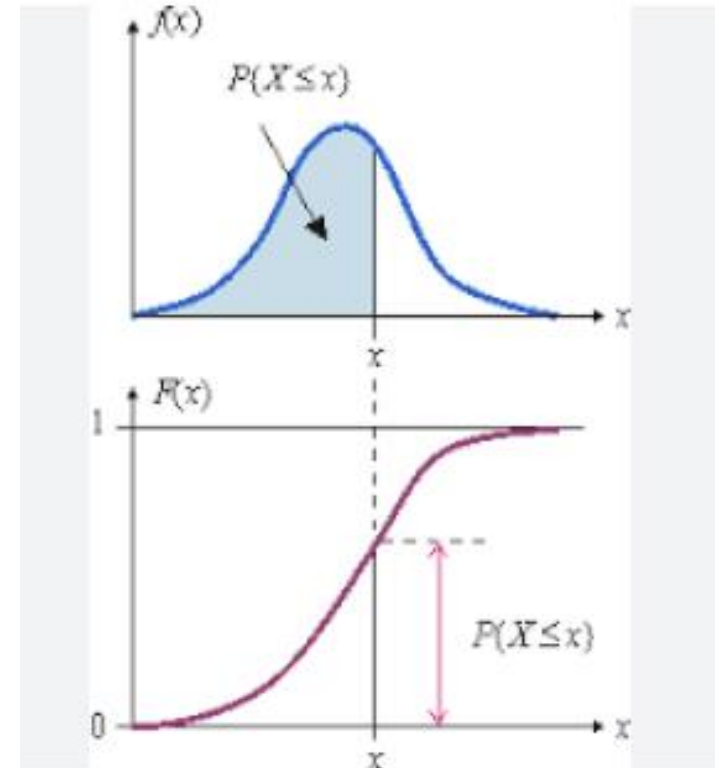
$$F_X(x) = P(X \leq x)$$

$F_X(x)$ = function of X

X = real value variable

P = probability that X will have a value less than or equal to x

$$F_X(x) = \int_{-\infty}^x f_X(t) dt. \quad \longleftrightarrow \quad f(x) = \frac{dF(x)}{dx}$$

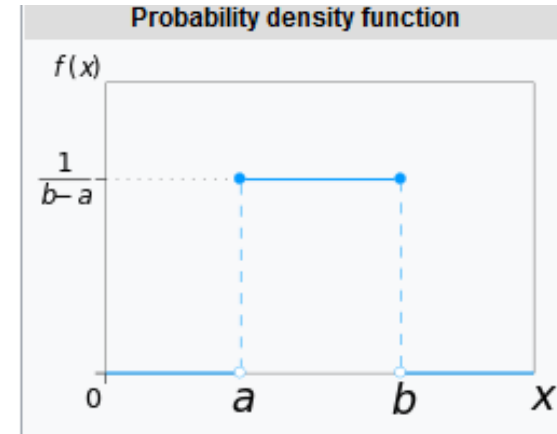


uniform distribution

Probability density function [\[edit \]](#)

The [probability density function](#) of the continuous uniform distribution is:

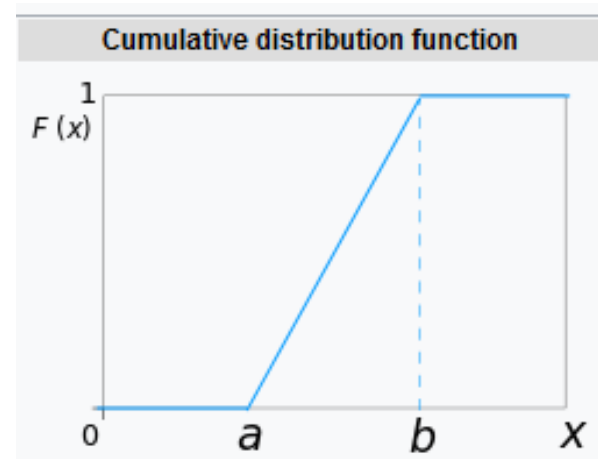
$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b. \end{cases}$$



Cumulative distribution function [\[edit \]](#)

The [cumulative distribution function](#) of the continuous uniform distribution is:

$$F(x) = \begin{cases} 0 & \text{for } x < a, \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b, \\ 1 & \text{for } x > b. \end{cases}$$

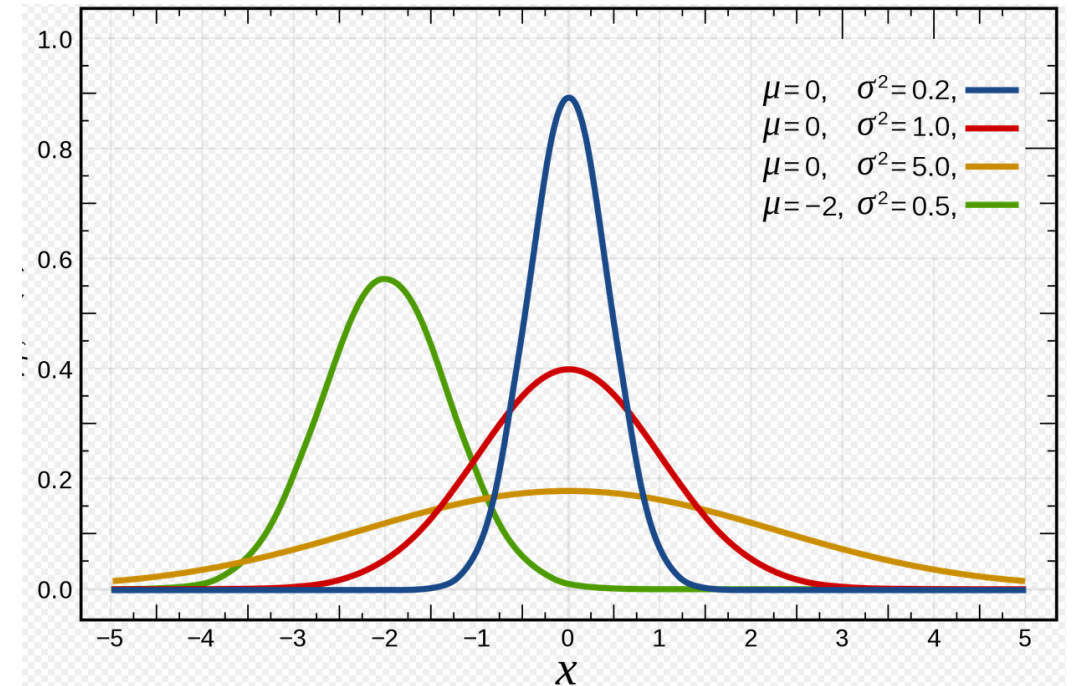
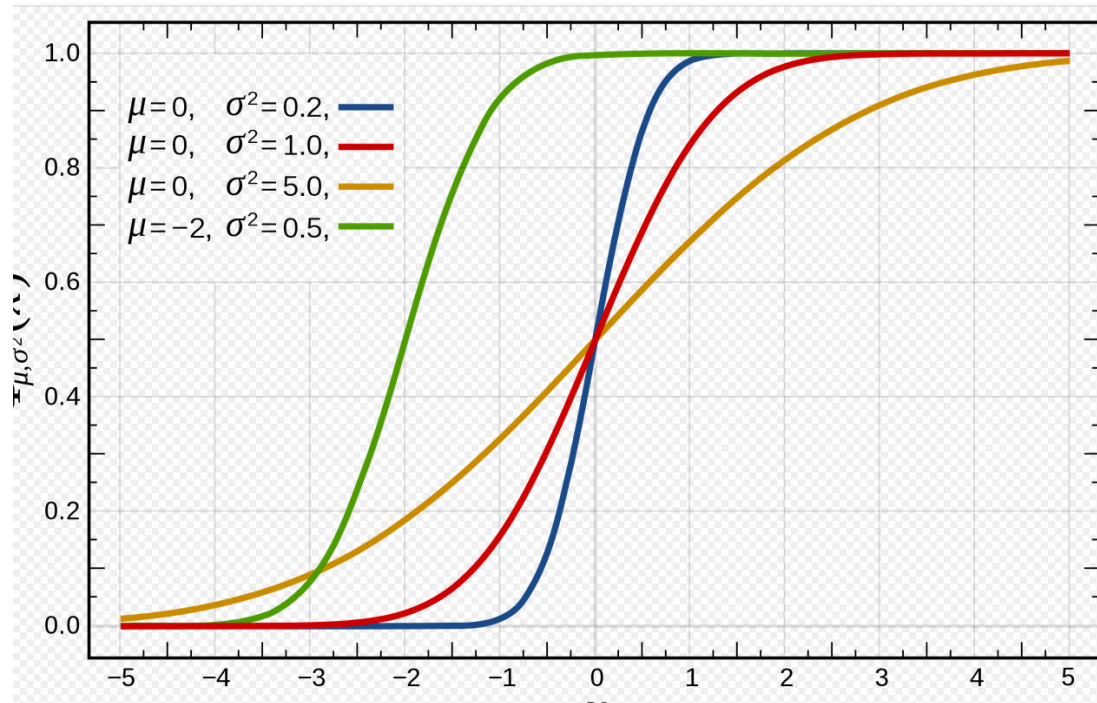
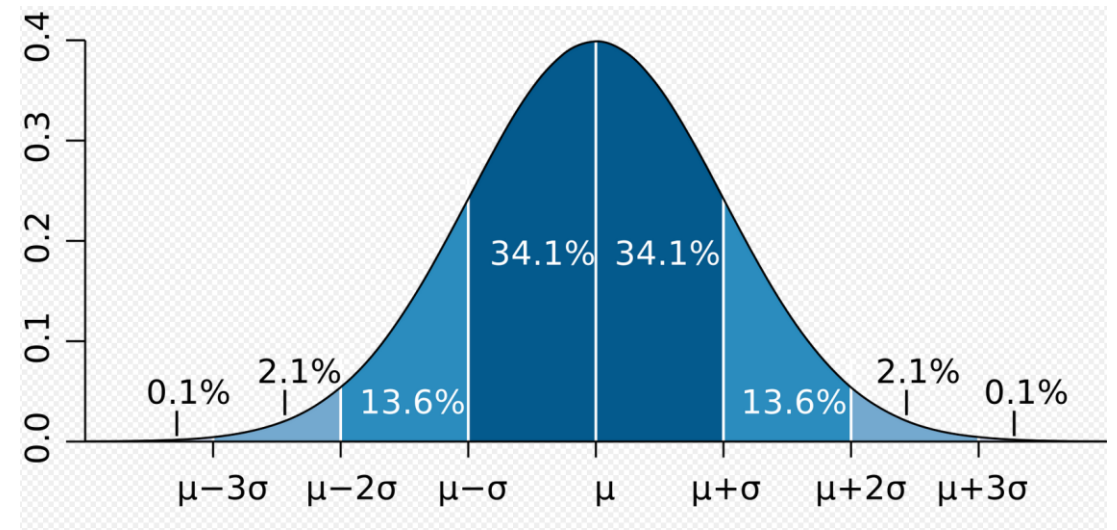


https://en.wikipedia.org/wiki/Continuous_uniform_distribution

In statistics, a **normal distribution** or **Gaussian distribution** is a type of **continuous probability distribution** for a **real-valued random variable**. The general form of its **probability density function** is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

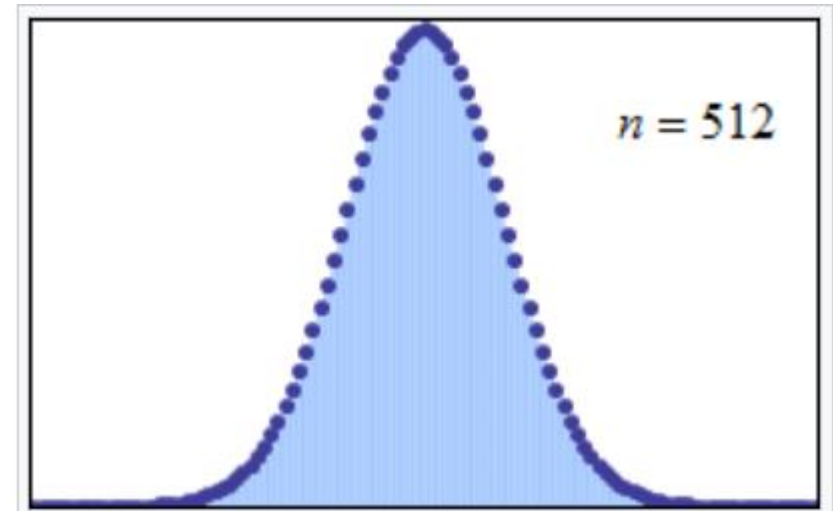
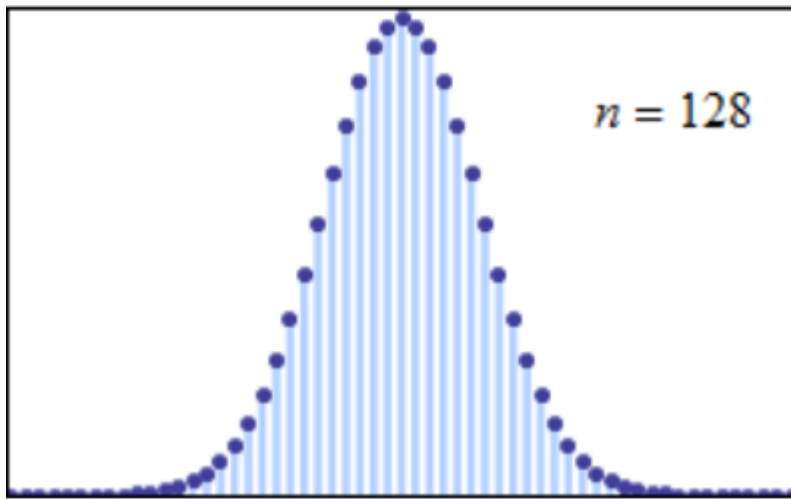
The parameter μ is the **mean** or **expectation** of the distribution (and also its **median** and **mode**), while the parameter σ is its **standard deviation**. The **variance** of the distribution is σ^2 . A random variable with a Gaussian distribution is said to be **normally distributed**, and is called a **normal deviate**.



https://en.wikipedia.org/wiki/Normal_distribution

De Moivre–Laplace theorem

$$\binom{n}{k} p^k q^{n-k} \simeq \frac{1}{\sqrt{2\pi npq}} e^{-\frac{(k-np)^2}{2npq}}, \quad p + q = 1, \quad p, q > 0$$



Exponential (Laplacian)

The exponential distribution

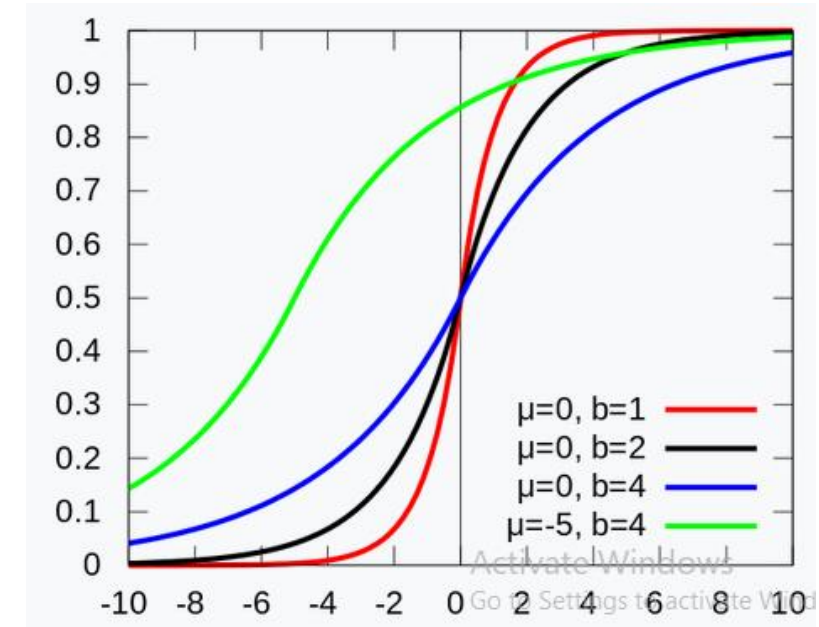
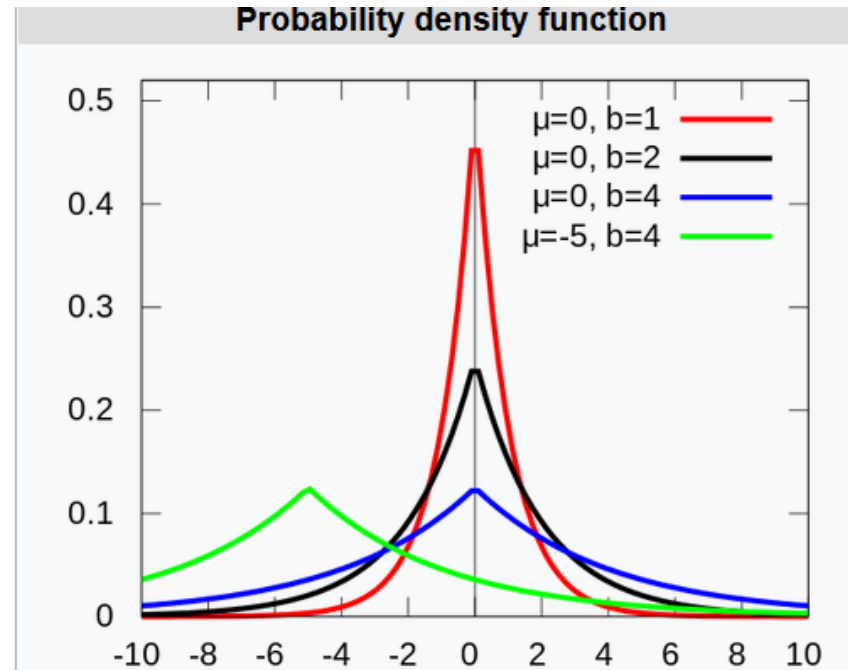
$$p(x) = \lambda e^{-\lambda x}, \quad x \geq 0 \quad (18)$$

is often used to describe the amount of time one must wait before an event occurs (such as an earthquake). In its two-sided form,

$$p(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \quad -\infty \leq x \leq \infty \quad (19)$$

it is known as the Laplacian and is often used to model natural image statistics.

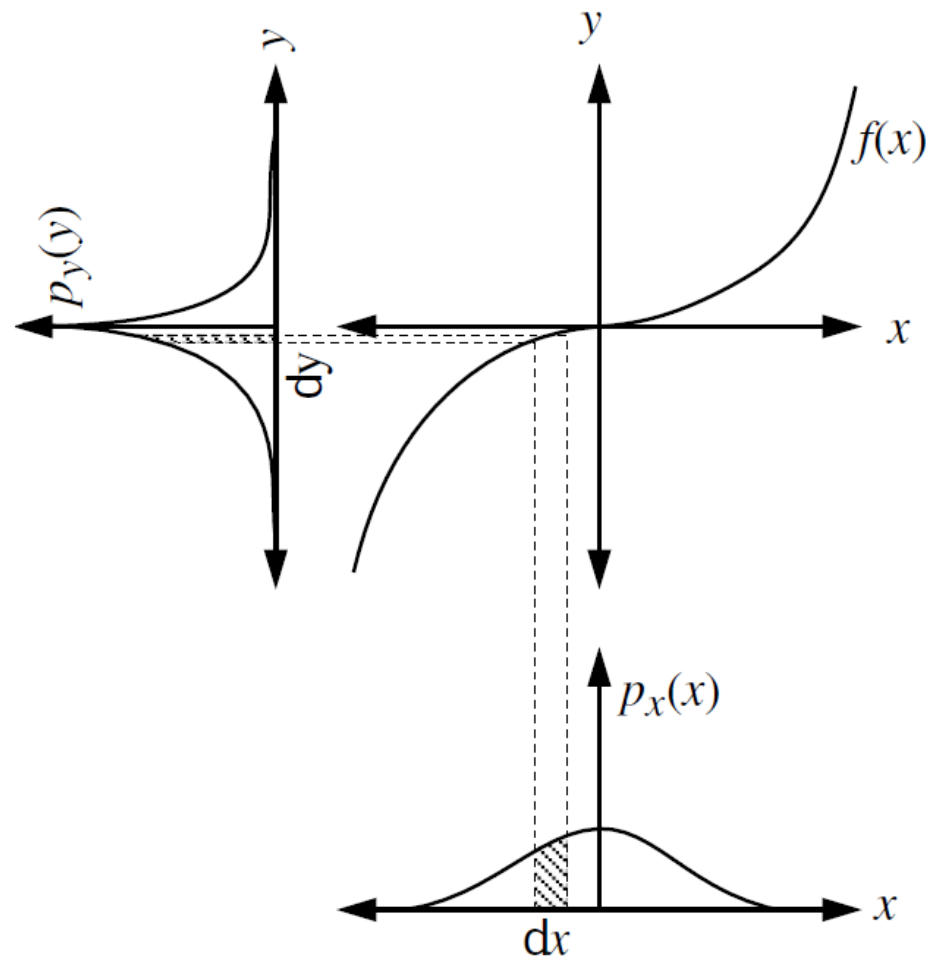
$$f(x | \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right)$$



Let's say we have a random variable x with distribution $p_x(x)$.

$$y = f(x)$$

what is the corresponding distribution of $p_y(y)$?



the area under the distribution must be preserved

$$\int_{x_0 - \frac{\delta x}{2}}^{x_0 + \frac{\delta x}{2}} p_x(x) dx = \int_{f(x_0 - \frac{\delta x}{2})}^{f(x_0 + \frac{\delta x}{2})} p_y(y) dy$$

$$p_x(x_0) \delta x = p_y(f(x_0)) \delta f(x_0)$$

$$p_y(y) = p_x(x) \left[\frac{dy}{dx} \right]^{-1}$$

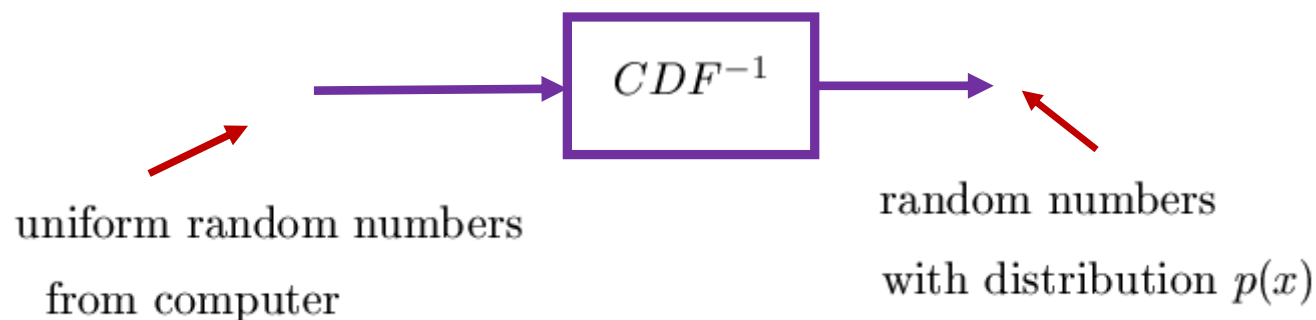
Generating random variables

y uniform distribution on $[0, 1]$ $\Rightarrow p_y(y) = 1$

Desired distribution : $p(x)$ $\Rightarrow y = f(x)? \Rightarrow x = f^{-1}(y)$

$$1 = p_x(x)[f']^{-1} \Rightarrow f'(x) = p(x) \Rightarrow$$

$$f(x) = \int_{-\infty}^x p_x(u) du$$



Moments

Mean (first moment)

$$\begin{aligned}\mu &= E[x] \\ &= \int p(x) x dx\end{aligned}$$

Variance (second moment)

$$\begin{aligned}\sigma^2 &= E[(x - \mu)^2] \\ &= \int p(x) (x - \mu)^2 dx\end{aligned}$$

$$\text{Standard deviation} \quad \sigma = \sqrt{\text{Variance}}$$

The mean (first raw **moment**)
of the continuous uniform distribution

$$\begin{aligned}E(X) &= \int_a^b x \frac{dx}{b-a} = \frac{b^2 - a^2}{2(b-a)} \\ &= \frac{a+b}{2}\end{aligned}$$

The variance (second **central moment**) of this distribution is:

$$\begin{aligned}V(X) &= E\left((X - E(X))^2\right) \\ &= \int_a^b \left(x - \frac{a+b}{2}\right)^2 \frac{dx}{b-a} = \frac{(b-a)^2}{12}.\end{aligned}$$

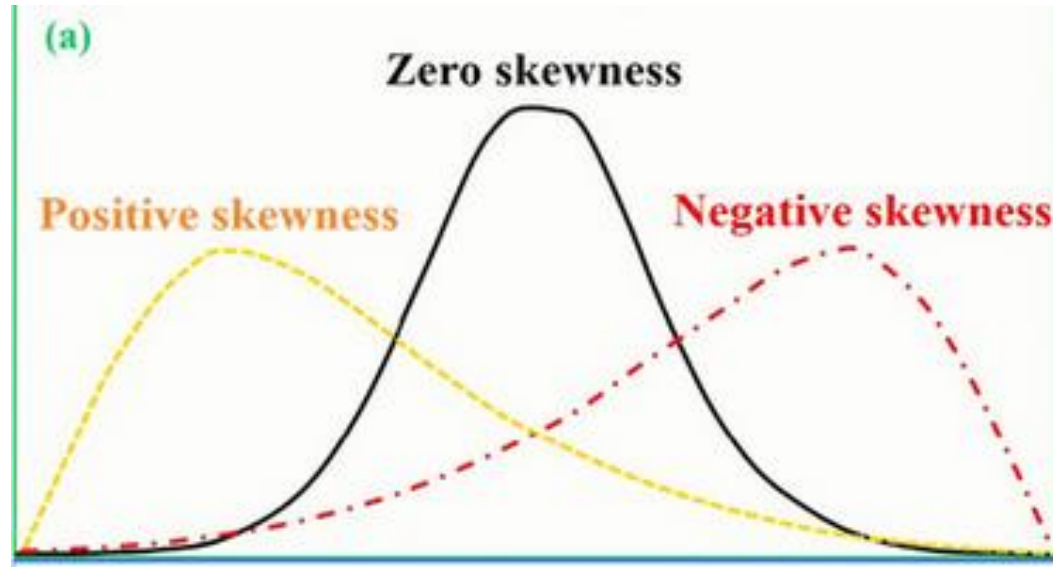
Skew (third moment)

The skew of a distribution attempts to characterize its lopsidedness:

$$\text{skew} = \frac{1}{\sigma^3} E[(x - \mu)^3] \quad (30)$$

$$= \frac{1}{\sigma^3} \int p(x) (x - \mu)^3 dx . \quad (31)$$

If the distribution is perfectly symmetric then the skew will be zero. But simply observing that the skew is zero does not necessarily imply the distribution is symmetric.



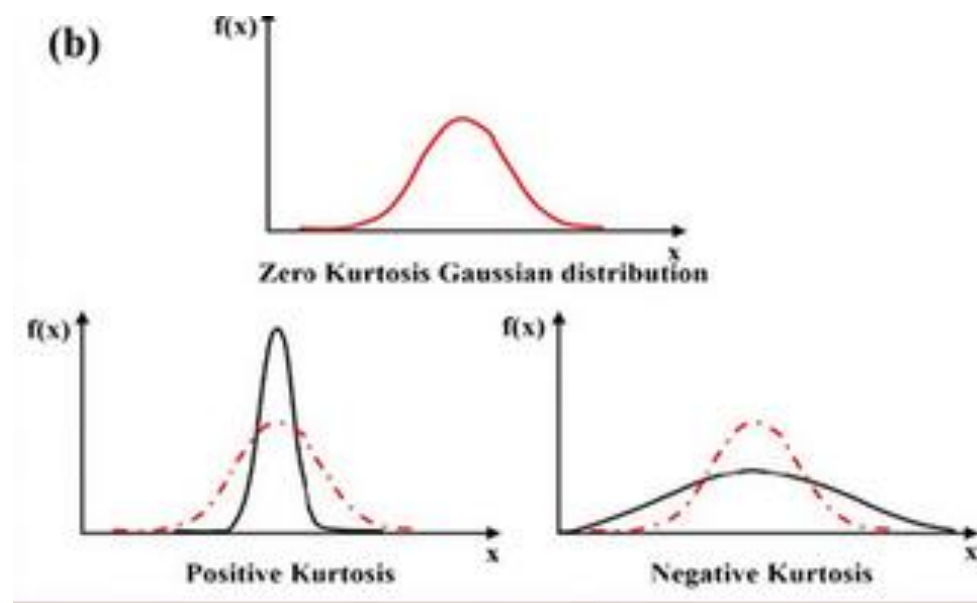
Kurtosis (proportional to fourth moment)

The kurtosis attempts to measure the peakedness of a distribution:

$$\kappa = \frac{1}{\sigma^4} E[(x - \mu)^4] - 3 \quad (32)$$

$$= \frac{1}{\sigma^4} \int p(x) (x - \mu)^4 dx - 3. \quad (33)$$

The reason for subtracting three is so that $\kappa = 0$ for a Gaussian distribution. A distribution with positive kurtosis may be more peaked or have heavier tails than a Gaussian, while a distribution with negative kurtosis may look more like a loaf of bread.



Law of large numbers

In **probability theory**, the **law of large numbers (LLN)** is a **theorem** that describes the result of performing the same experiment a large number of times. According to the law, the **average** of the results obtained from a large number of trials should be close to the **expected value** and tends to become closer to the expected value as more trials are performed.^[1]

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{X_i}{n} = \overline{X}$$

$$E(X_1) = E(X_2) = \dots = \mu, \quad \text{Var}(X_i) = \sigma^2$$

$$\overline{X}_n \rightarrow \mu \quad \text{as } n \rightarrow \infty.$$

$$\text{Var}(\overline{X}_n) = \text{Var}\left(\frac{1}{n}(X_1 + \dots + X_n)\right) = \frac{1}{n^2} \text{Var}(X_1 + \dots + X_n) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

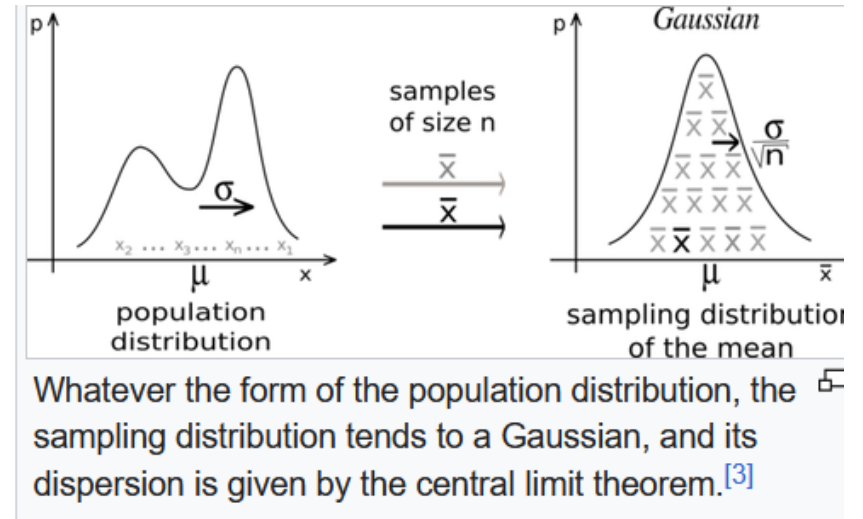
https://en.wikipedia.org/wiki/Law_of_large_numbers

Central limit theorem

In probability theory, the **central limit theorem (CLT)** establishes that, in many situations, for independent and identically distributed random variables, the sampling distribution of the standardized sample mean tends towards the standard normal distribution even if the original variables themselves are not normally distributed.

Let $\{X_1, \dots, X_n\}$ be a sequence of i.i.d. random variables having a distribution with expected value given by μ and finite variance given by σ^2 . Suppose we are interested in the sample average

$$\bar{X}_n \equiv \frac{X_1 + \dots + X_n}{n}.$$



Lindeberg–Lévy CLT — Suppose $\{X_1, \dots, X_n\}$ is a sequence of i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$. Then, as n approaches infinity, the random variables $\sqrt{n}(\bar{X}_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$:^[4]

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

Principle of maximum entropy

$$H(x) = - \sum_i x_i \log(x_i)$$

$$H(x) = E(-\ln(p(x))) = - \int p(x) \ln(p(x)) dx$$

The **principle of maximum entropy** states that the **probability distribution** which best represents the current state of knowledge about a system is the one with largest **entropy**, in the context of precisely stated prior data (such as a **proposition** that expresses **testable information**).

https://en.wikipedia.org/wiki/Principle_of_maximum_entropy

Bernoulli

A Bernoulli random variable is simply a binary variable (i.e., it occupies one of two states) with

$$P(1) = p \quad (4)$$

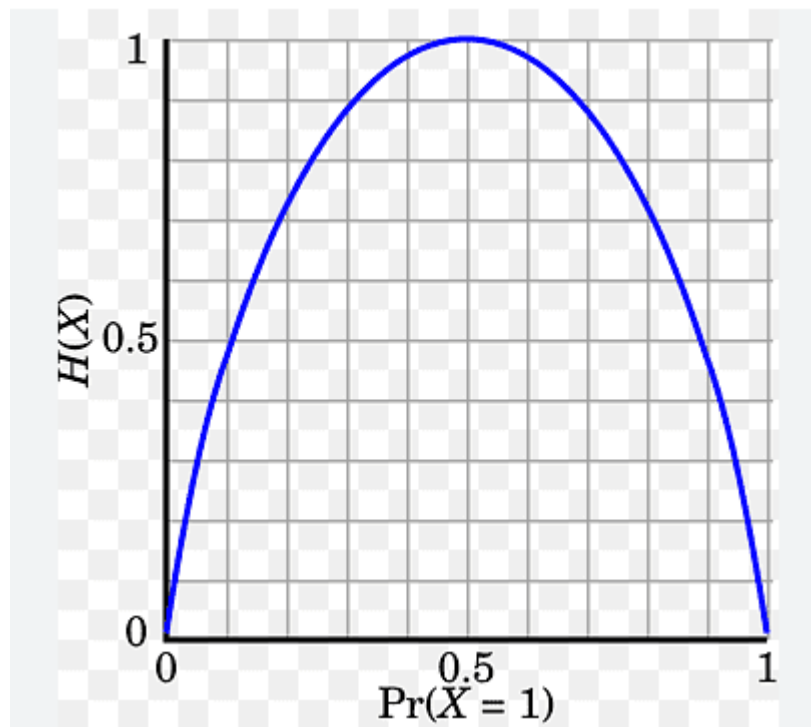
$$P(0) = 1 - p \quad (5)$$

$$H(p) = -p \log(p) - (1 - p) \log(1 - p)$$

$$\max_p H(p)$$

$$H(p) = -p \log(p) - (1-p) \log(1-p)$$

$$\frac{dH}{dp} = \log \frac{1-p}{p} = 0 \quad \Rightarrow \quad \frac{1-p}{p} = 1 \quad \Rightarrow \quad p = \frac{1}{2}$$



uniform distribution

More generally, we may take the expected value of any arbitrary function $\phi(x)$ in an attempt to characterize a distribution:

$$\alpha = E[\phi(x)] \tag{34}$$

$$= \int p(x) \phi(x) dx . \tag{35}$$

$$p(x) = \frac{1}{Z_\lambda} e^{-\lambda \phi(x)}$$

https://en.wikipedia.org/wiki/Maximum_entropy_probability_distribution

Covariance

$$\begin{aligned}
 \text{cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\
 &= E[XY - X E[Y] - E[X] Y + E[X] E[Y]] \\
 &= E[XY] - E[X] E[Y] - E[X] E[Y] + E[X] E[Y] \\
 &= E[XY] - E[X] E[Y],
 \end{aligned}$$

$$\text{cov}(X, X) = \text{var}(X) \equiv \sigma^2(X) \equiv \sigma_X^2.$$

Suppose that X and Y have the following [joint probability mass function](#),^[6] in which the six central cells give the discrete joint probabilities $f(x, y)$ of the six hypothetical realizations $(x, y) \in S = \{(5, 8), (6, 8), (7, 8), (5, 9), (6, 9), (7, 9)\}$:

$f(x, y)$		x			$f_Y(y)$
		5	6	7	
y	8	0	0.4	0.1	0.5
	9	0.3	0	0.2	0.5
$f_X(x)$		0.3	0.4	0.3	1

$$f_Y(y) = \sum_x P(x, y)$$

$$f_X(x) = \sum_y P(x, y)$$

$$\mu_X = 5(0.3) + 6(0.4) + 7(0.1 + 0.2) = 6$$

$$\mu_Y = 8(0.4 + 0.1) + 9(0.3 + 0.2) = 8.5$$

$$\sigma_X^2 = \sum_i [x_i(x_i - \mu_X)^2]$$

$$\sigma_Y^2 = \sum_j [y_j(y_j - \mu_Y)^2]$$

$$\begin{aligned}
 \text{cov}(X, Y) &= \sigma_{XY} = \sum_{(x,y) \in S} f(x, y) (x - \mu_X) (y - \mu_Y) \\
 &= (0)(5 - 6)(8 - 8.5) + (0.4)(6 - 6)(8 - 8.5) + (0.1)(7 - 6)(8 - 8.5) + \\
 &\quad (0.3)(5 - 6)(9 - 8.5) + (0)(6 - 6)(9 - 8.5) + (0.2)(7 - 6)(9 - 8.5) \\
 &= -0.1 .
 \end{aligned}$$

<https://en.wikipedia.org/wiki/Covariance>

Correlation

$$\rho_{X,Y} = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}$$

$$\begin{aligned} X, Y \text{ independent} &\Rightarrow \rho_{X,Y} = 0 \quad (X, Y \text{ uncorrelated}) \\ \rho_{X,Y} = 0 \quad (X, Y \text{ uncorrelated}) &\nRightarrow X, Y \text{ independent} \end{aligned}$$

For example, suppose the random variable X is symmetrically distributed about zero, and $Y = X^2$. Then Y is completely determined by X , so that X and Y are perfectly dependent, but their correlation is zero; they are [uncorrelated](#). However, in the special case when X and Y are [jointly normal](#), uncorrelatedness is equivalent to independence.

<https://en.wikipedia.org/wiki/Correlation>